

SYNOPSIS

DATAPILOT AI: AUTONOMOUS BUSINESS INTELLIGENCE & DECISION INTELLIGENCE

ABSTRACT / EXECUTIVE SUMMARY

Traditional business analytics infrastructures are typically isolated into disconnected components: raw transactional databases, data warehouses, analytical charting engines, and machine learning scoring environments. This project introduces DataPilot AI, an autonomous business intelligence and decision intelligence platform designed to collapse these boundaries. DataPilot AI unifies data ingestion, polyglot storage layer abstractions, real-time streaming, explainable machine learning pipelines, and autonomous multi-agent systems into a single decision-support platform. The system uses a unified Model Context Protocol (MCP) interface to decouple data tools from large language models, coordinating intelligent workflows through a structured multi-agent pipeline of Planner, Executor, and Analyzer agents. Built with LangGraph, DuckDB, Streamlit, and Neo4j, the platform dynamically processes natural-language queries and delivers real-time forecasting, anomaly tracking, and proactive alert routing.

INTRODUCTION

The domain of enterprise business intelligence is undergoing a significant shift driven by autonomous multi-agent AI systems and standardized model-tool interfaces. Industry trends emphasize a transition from static, human-driven dashboard reporting to dynamic, autonomous data investigation. A majority of organizations are seeking to integrate intelligent automation within their core data pipelines to accelerate decision-making. By incorporating the emerging Model Context Protocol standard alongside explainable machine learning workflows, modern decision intelligence platforms can securely bridge raw data layers with automated reasoning and communication channels.

PROBLEM STATEMENT

Modern data architectures suffer from fragmentation, where transactional systems, analytical warehouses, and predictive machine learning environments remain isolated from one another. This separation requires complex, manual orchestration to extract actionable business insights from multiple independent layers. Standard language-model integrations also introduce data exposure risks and lack transparency, making automated decisions difficult to verify. Current frameworks fail to provide a unified, secure interface capable of translating natural language queries into safe, multi-database executions while generating explainable analytical output.

OBJECTIVES

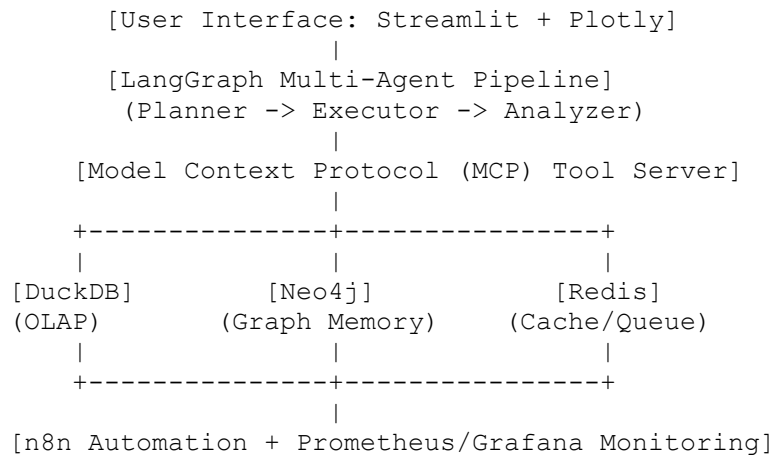
- To implement a secure natural language to optimized query processing engine using a multi-agent LangGraph pipeline.
- To establish a structured Planner → Executor → Analyzer agent workflow that isolates each agent's responsibilities.
- To develop a Model Context Protocol (MCP) tool layer that decouples database adapters from the orchestration layer.
- To build an interactive Streamlit and Plotly dashboard for real-time visualization of analytical results.
- To integrate workflow automation and monitoring using n8n, Redis, Prometheus, and Grafana for alerting and observability.

PROPOSED METHODOLOGY / SYSTEM DESIGN

The platform implements a three-agent workflow orchestrated using LangGraph state machines. The Planner agent interprets natural language queries and decomposes them into executable analytical steps; the Executor agent runs the corresponding data operations against the storage layer; and the Analyzer agent synthesizes results into human-readable insights and visualizations. The frontend is built using Streamlit with Plotly for interactive charting, while DuckDB serves as the primary analytical query engine for fast, in-process OLAP computation. Neo4j is used to

maintain semantic graph memory across agent interactions, and Redis supports caching and inter-agent messaging. The project follows an agile development approach across a six-month build plan, with an initial study phase prioritizing LangGraph, DuckDB, and MCP.

System Architecture (Block Diagram)



HARDWARE / SOFTWARE SPECIFICATION

Specification Type	Selected Technology Components
Software Environment	Python 3.11+, LangGraph, DuckDB, Streamlit, Plotly, Neo4j, Redis, n8n, Prometheus, Grafana, MCP
Hardware Environment	Intel i5/i7 Processor or equivalent, 16GB System RAM, 256GB Solid State Drive, Network Interface Card

CONCLUSION

DataPilot AI offers a practical alternative to fragmented business intelligence tools by combining multi-agent orchestration, the Model Context Protocol, and explainable analytics into a single decision intelligence platform, enabling faster and more transparent data-driven decisions.

FUTURE SCOPE

- Can be extended to small and mid-sized enterprises for self-service analytics without dedicated data teams.

- Integration with role-based access control and enterprise single sign-on for secure multi-user deployment.
- Expansion of the agent pipeline to support distributed, multi-node local LLM inference.

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